

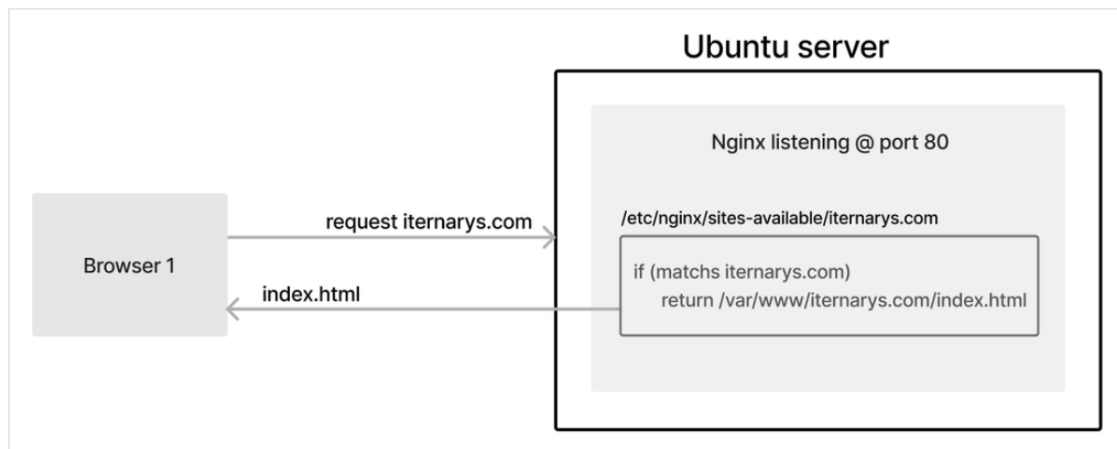


## Guia 3 – Web server

### Objetivos:

- Instalar um Servidor Web usando a ferramenta NGINX

Diagrama de funcionamento:



## NGINX web server

Instalar NGINX.

```
sudo apt update  
sudo apt install nginx
```

Verificar o estado do servidor.

```
systemctl status nginx
```

nginx.service - A high performance web server and a reverse proxy server

Loaded: loaded (/lib/systemd/system/nginx.service; enabled; vendor preset: enabled)

Active: active (running) since Fri 2020-04-20 16:08:19 UTC; 3 days ago

Docs: man:nginx(8)

Main PID: 2369 (nginx)

Tasks: 2 (limit: 1153)

Memory: 3.5M

CGroup: /system.slice/nginx.service

└─2369 nginx: master process /usr/sbin/nginx -g daemon on; master\_process on;

└─2380 nginx: worker process



Edite o ficheiro `/etc/hosts` e coloque a seguinte linha (no início do ficheiro):

```
127.0.0.1 iternerays.com
```

Nota: aprenderemos mais sobre estas instruções quando falarmos do protocolo DNS; com esta configuração poderá – no seu *browser* – colocar o nome `iternerays.com` que será resolvido para o IP 127.0.0.1

Leia com atenção e siga os passos que se encontram no seguinte tutorial

- <https://episyche.com/blog/how-to-set-up-a-virtual-host-on-an-nginx>

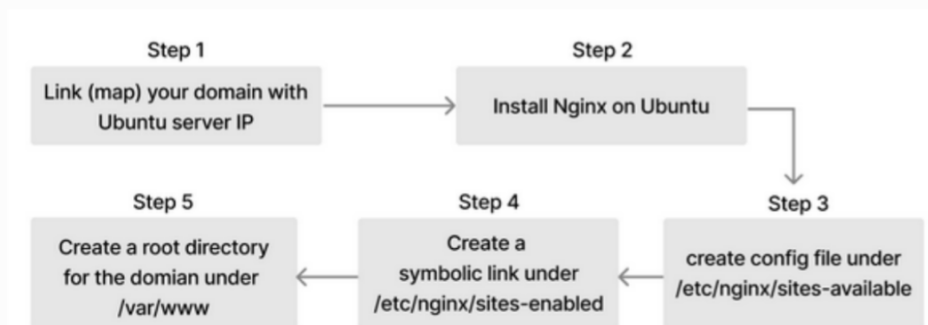
### Questões

1. Inicie uma captura de pacotes HTTP.
2. Aceda ao domínio `iternerays.com`
3. (sem parar a captura) Faça *refresh* da página.
4. Explique os pacotes HTTP que capturou, o seu significado e o seu conteúdo. Entregue as respostas via *eLearning*.

```
labi@labi-VirtualBox: ~  
labi@labi-VirtualBox:~$ systemctl status nginx  
● nginx.service - A high performance web server and a reverse proxy server  
   Loaded: loaded (/lib/systemd/system/nginx.service; enabled; vendor preset: enabled)  
   Active: active (running) since Fri 2025-10-03 11:37:59 WEST; 1min 47s ago  
     Docs: man:nginx(8)  
  Main PID: 3171 (nginx)  
    Tasks: 3 (limit: 4573)  
   Memory: 3.7M  
   CGroup: /system.slice/nginx.service  
           └─3171 nginx: master process /usr/sbin/nginx -g daemon on; master_process on;  
             └─3172 nginx: worker process  
               └─3173 nginx: worker process  
  
out 03 11:37:59 labi-VirtualBox systemd[1]: Starting A high performance web server and a reverse proxy server:  
out 03 11:37:59 labi-VirtualBox systemd[1]: Started A high performance web server and a reverse proxy server:  
lines 1-14/14 (END)
```

```
labi@labi-VirtualBox:/etc$ sudo nano hosts  
labi@labi-VirtualBox:/etc$ cat hosts  
127.0.0.1        iternerays.com  
127.0.0.1        localhost  
127.0.1.1        labi-VirtualBox  
  
# The following lines are desirable for IPv6 capable hosts  
::1             ip6-localhost ip6-loopback  
fe00::0         ip6-localnet  
ff00::0         ip6-mcastprefix  
ff02::1         ip6-allnodes  
ff02::2         ip6-allrouters
```

## Questões:



```

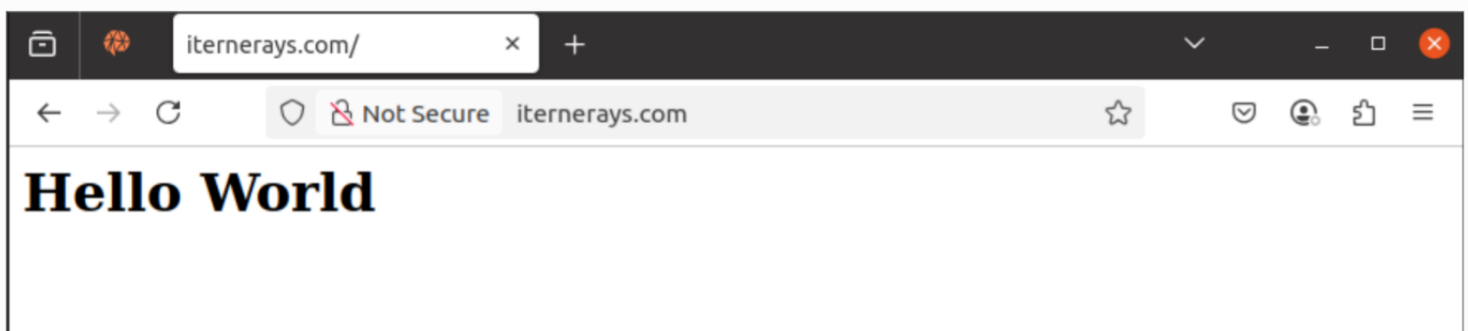
labi@labi-VirtualBox:~$ cd /etc/nginx/sites-available/
labi@labi-VirtualBox:/etc/nginx/sites-available$ ll
total 12
drwxr-xr-x 2 root root 4096 out  3 11:37 ./
drwxr-xr-x 8 root root 4096 out  3 11:37 ../
-rw-r--r-- 1 root root 2416 mar 20 2024 default
labi@labi-VirtualBox:/etc/nginx/sites-available$ sudo nano iternerays.com
[sudo] password for labi:
labi@labi-VirtualBox:/etc/nginx/sites-available$ ll
total 16
drwxr-xr-x 2 root root 4096 out  3 12:15 ./
drwxr-xr-x 8 root root 4096 out  3 11:37 ../
-rw-r--r-- 1 root root 2416 mar 20 2024 default
-rw-r--r-- 1 root root  223 out  3 12:15 iternerays.com
labi@labi-VirtualBox:/etc/nginx/sites-available$ cat iternerays.com
server {
    listen 80;
    listen [::]:80;

    server_name itinerarays.com;

    root /var/www/itinerarays.com;
    index index.html;

    location / {
        try_files $uri $uri/ =404;
    }
}
labi@labi-VirtualBox:/etc/nginx/sites-available$ sudo ln -s /etc/nginx/sites-available/iternerays.com /etc/nginx/sites-enabled/
labi@labi-VirtualBox:/etc/nginx/sites-available$ l
default iternerays.com
labi@labi-VirtualBox:/etc/nginx/sites-available$ ll
total 16
drwxr-xr-x 2 root root 4096 out  3 12:15 ./
drwxr-xr-x 8 root root 4096 out  3 11:37 ../
-rw-r--r-- 1 root root 2416 mar 20 2024 default
-rw-r--r-- 1 root root  223 out  3 12:15 iternerays.com
labi@labi-VirtualBox:/etc/nginx/sites-available$ cd ..
labi@labi-VirtualBox:/etc/nginx$ cd sites-enabled/
labi@labi-VirtualBox:/etc/nginx/sites-enabled$ ll
total 8
drwxr-xr-x 2 root root 4096 out  3 12:16 ./
drwxr-xr-x 8 root root 4096 out  3 11:37 ../
lrwxrwxrwx 1 root root  34 out  3 11:37 default -> /etc/nginx/sites-available/default
lrwxrwxrwx 1 root root  41 out  3 12:16 iternerays.com -> /etc/nginx/sites-available/iternerays.com

```



1) + 2) + 3) :

iternerays.com/

Not Secure iternerays.com

# Hello World

labi@labi-VirtualBox: ~  
labi@labi-VirtualBox:~\$ sudo systemctl stop nginx  
[sudo] password for labi:  
labi@labi-VirtualBox:~\$ sudo systemctl start nginx  
labi@labi-VirtualBox:~\$ sudo systemctl status nginx  
● nginx.service - A high performance web server and a reverse proxy server  
 Loaded: loaded (/lib/systemd/system/nginx.service; enabled; vendor preset: enabled)  
 Active: active (running) since Fri 2025-10-03 15:56:53 WEST; 5s ago  
 Docs: man:nginx(8)  
 Process: 2658 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master\_process on; (code=exited, status=0/SUCCESS)  
 Process: 2659 ExecStart=/usr/sbin/nginx -g daemon on; master\_process on; (code=exited, status=0/SUCCESS)  
 Main PID: 2660 (nginx)  
 Tasks: 3 (limit: 4573)  
 Memory: 3.1M  
 CGroup: /system.slice/nginx.service  
 └─2660 nginx: master process /usr/sbin/nginx -g daemon on; master\_process on;  
 └─2661 nginx: worker process  
 └─2662 nginx: worker process  
  
out 03 15:56:53 labi-VirtualBox systemd[1]: Starting A high performance web server: nginx.service.  
out 03 15:56:53 labi-VirtualBox systemd[1]: Started A high performance web server: nginx.service.  
lines 1-16/16 (END)

\*Loopback: lo

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

http

| No. | Time         | Source    | Destination | Protocol | Length | Info                               |
|-----|--------------|-----------|-------------|----------|--------|------------------------------------|
| 393 | 11.038849204 | 127.0.0.1 | 127.0.0.1   | HTTP     | 373    | HTTP/1.1 200 OK (text/html)        |
| 395 | 11.704672614 | 127.0.0.1 | 127.0.0.1   | HTTP     | 425    | GET /favicon.ico HTTP/1.1          |
| 396 | 11.706332891 | 127.0.0.1 | 127.0.0.1   | HTTP     | 395    | HTTP/1.1 404 Not Found (text/html) |
| 398 | 15.202697908 | 127.0.0.1 | 127.0.0.1   | HTTP     | 499    | GET / HTTP/1.1                     |
| 399 | 15.202953094 | 127.0.0.1 | 127.0.0.1   | HTTP     | 254    | HTTP/1.1 304 Not Modified          |

Frame 391: 408 bytes on wire (3264 bits), 408 bytes captured (3264 bits) on interface lo, id 0  
Ethernet II, Src: 00:00:00:00:00:00 (00:00:00:00:00:00), Dst: 00:00:00:00:00:00 (00:00:00:00:00:00)  
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1  
Transmission Control Protocol, Src Port: 56102, Dst Port: 80, Seq: 1, Ack: 1, Len: 342  
Hypertext Transfer Protocol  
GET / HTTP/1.1  
[Expert Info (Chat/Sequence): GET / HTTP/1.1  
[GET / HTTP/1.1  
[Severity level: Chat]  
[Group: Sequence]  
Request Method: GET  
Request URI: /  
Request Version: HTTP/1.1  
User-Agent: Mozilla/5.0 (X11; Ubuntu; Linux x86\_64; rv:136.0) Gecko/20100101 Firefox/136.0  
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,\*/\*;q=0.8  
Accept-Language: en-US,en;q=0.5  
Accept-Encoding: gzip, deflate  
Connection: keep-alive  
Upgrade-Insecure-Requests: 1  
Priority: u=0, i=1  
Full request URI: http://iternerays.com/  
HTTP request 1/3  
Response in frame: 393  
Next request in frame: 395

0050 0d 0a 4b 8f 73 74 3a 20 69 74 65 72 6e 65 72 63 --Host: iternerays.com  
0060 f9 73 2e 63 ef 6d 0d 0a 55 73 65 72 2d 41 67 65 .s.com: User-Agent  
0070 6e 74 3a 20 4d 6f 7a 69 6c 6c 61 2f 35 2e 30 20 nt: Mozilla/5.0  
0080 28 58 31 31 3b 20 55 62 75 6e 74 75 3b 20 4c 69 (X11; Ubuntu; Linux x86\_64; rv:136.0) Gecko/20100101 Firefox/136.0  
0090 6e 75 78 20 78 3b 36 5f 36 34 3b 20 72 76 3a 31 nux x86\_64; rv:136.0) Gecko/20100101 Firefox/136.0  
00a0 33 36 2e 30 29 20 47 65 63 6b 6f 2f 32 30 31 30 36.0) Gecko/20100101 Firefox/136.0  
00b0 30 31 30 31 20 46 69 72 65 66 6f 78 2f 31 33 36 0101 Firefox/136.0

HTTP Host (http.host). 22 bytes

Packets: 451 · Displayed: 6 (1.3%)

Profile: Default



Wireshark interface showing a packet capture on interface lo. The selected packet is a 404 Not Found response from 127.0.0.1 to 127.0.0.1. The packet details pane shows the Hypertext Transfer Protocol section expanded, displaying the 404 status and the response body (123 bytes). The packet bytes pane shows the raw data of the response body.

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

http

| No. | Time          | Source    | Destination | Protocol | Length | Info                               |
|-----|---------------|-----------|-------------|----------|--------|------------------------------------|
| 391 | 111.627264696 | 127.0.0.1 | 127.0.0.1   | HTTP     | 408    | GET / HTTP/1.1                     |
| 393 | 111.638849284 | 127.0.0.1 | 127.0.0.1   | HTTP     | 373    | HTTP/1.1 200 OK (text/html)        |
| 395 | 111.704672614 | 127.0.0.1 | 127.0.0.1   | HTTP     | 425    | GET /favicon.ico HTTP/1.1          |
| 396 | 111.706332891 | 127.0.0.1 | 127.0.0.1   | HTTP     | 395    | HTTP/1.1 404 Not Found (text/html) |
| 398 | 115.202697908 | 127.0.0.1 | 127.0.0.1   | HTTP     | 490    | GET / HTTP/1.1                     |
| 399 | 115.202953094 | 127.0.0.1 | 127.0.0.1   | HTTP     | 254    | HTTP/1.1 304 Not Modified          |

Frame 396: 395 bytes on wire (3160 bits), 395 bytes captured (3160 bits) on interface lo, id 0

Ethernet II, Src: 00:00:00:00:00:00 (00:00:00:00:00:00), Dst: 00:00:00:00:00:00 (00:00:00:00:00:00)

Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1

Transmission Control Protocol, Src Port: 80, Dst Port: 56102, Seq: 308, Ack: 702, Len: 329

Hypertext Transfer Protocol

HTTP/1.1 404 Not Found

[Expert Info (Chat/Sequence): HTTP/1.1 404 Not Found]

[HTTP/1.1 404 Not Found]

[Severity level: Chat]

[Group: Sequence]

Response Version: HTTP/1.1

Status Code: 404

[Status Code Description: Not Found]

Response Phrase: Not Found

Server: nginx/1.18.0 (Ubuntu)

Date: Fri, 03 Oct 2025 15:09:20 GMT

Content-Type: text/html

Transfer-Encoding: chunked

Connection: keep-alive

Content-Encoding: gzip

HTTP response 2/3

[Time since request: 0.001660277 seconds]

[Prev request in frame: 391]

[Prev response in frame: 393]

[Request in frame: 395]

[Next request in frame: 398]

[Next response in frame: 399]

[Request URI: http://iternerays.com/]

HTTP chunked response

Content-encoded entity body (gzip): 123 bytes -> 162 bytes

File Data: 162 bytes

Line-based text data: text/html (7 lines)

<html>

<head><title>404 Not Found</title></head>

<body>

<center><h1>404 Not Found</h1></center>

<hr><center>nginx/1.18.0 (Ubuntu)</center>

</body>

</html>

Frame 395 (395 bytes) De-chunked entity body (123 bytes) Uncompressed entity body (162 bytes)

Hypertext Transfer Protocol: Protocol Packets: 495 · Displayed: 6 (1.2%) Profile: Default

Wireshark interface showing a packet capture on interface lo. The selected packet is a GET request from 127.0.0.1 to 127.0.0.1. The packet details pane shows the Hypertext Transfer Protocol section expanded, displaying the GET method and the request body (22 bytes). The packet bytes pane shows the raw data of the request body.

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

http

| No. | Time          | Source    | Destination | Protocol | Length | Info                               |
|-----|---------------|-----------|-------------|----------|--------|------------------------------------|
| 391 | 111.627264696 | 127.0.0.1 | 127.0.0.1   | HTTP     | 408    | GET / HTTP/1.1                     |
| 393 | 111.638849284 | 127.0.0.1 | 127.0.0.1   | HTTP     | 373    | HTTP/1.1 200 OK (text/html)        |
| 395 | 111.704672614 | 127.0.0.1 | 127.0.0.1   | HTTP     | 425    | GET /favicon.ico HTTP/1.1          |
| 396 | 111.706332891 | 127.0.0.1 | 127.0.0.1   | HTTP     | 395    | HTTP/1.1 404 Not Found (text/html) |
| 398 | 115.202697908 | 127.0.0.1 | 127.0.0.1   | HTTP     | 490    | GET / HTTP/1.1                     |
| 399 | 115.202953094 | 127.0.0.1 | 127.0.0.1   | HTTP     | 254    | HTTP/1.1 304 Not Modified          |

Frame 398: 490 bytes on wire (3920 bits), 490 bytes captured (3920 bits) on interface lo, id 0

Ethernet II, Src: 00:00:00:00:00:00 (00:00:00:00:00:00), Dst: 00:00:00:00:00:00 (00:00:00:00:00:00)

Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1

Transmission Control Protocol, Src Port: 56102, Dst Port: 80, Seq: 702, Ack: 637, Len: 424

Hypertext Transfer Protocol

GET / HTTP/1.1

[Expert Info (Chat/Sequence): GET / HTTP/1.1]

[GET / HTTP/1.1]

[Severity level: Chat]

[Group: Sequence]

Request Method: GET

Request URI: /

Request Version: HTTP/1.1

Host: iternerays.com

User-Agent: Mozilla/5.0 (X11; Ubuntu; Linux x86\_64; rv:136.0) Gecko/20100101 Firefox/136.0

Accept: text/html,application/xhtml+xml,application/xml;q=0.9,\*/\*;q=0.8

Accept-Language: en-US,en;q=0.5

Accept-Encoding: gzip, deflate

Connection: keep-alive

Upgrade-Insecure-Requests: 1

If-Modified-Since: Fri, 03 Oct 2025 11:22:05 GMT

If-None-Match: W/"68dfb1d0-15"

Priority: u=0, i=1

Full request URI: http://iternerays.com/

HTTP request 3/3

[Prev request in frame: 395]

[Response in frame: 399]

Frame 398 (490 bytes) HTTP Host (http.host). 22 bytes Packets: 512 · Displayed: 6 (1.2%) Profile: Default

\*Loopback: lo

FileEditViewGoCaptureAnalyzeStatisticsTelephonyWirelessToolsHelp

http

| No. | Time          | Source    | Destination | Protocol | Length | Info                               |
|-----|---------------|-----------|-------------|----------|--------|------------------------------------|
| 391 | 111.627264696 | 127.0.0.1 | 127.0.0.1   | HTTP     | 468    | GET / HTTP/1.1                     |
| 393 | 111.638849294 | 127.0.0.1 | 127.0.0.1   | HTTP     | 373    | HTTP/1.1 200 OK (text/html)        |
| 395 | 111.704672614 | 127.0.0.1 | 127.0.0.1   | HTTP     | 425    | GET /favicon.ico HTTP/1.1          |
| 396 | 111.706332891 | 127.0.0.1 | 127.0.0.1   | HTTP     | 395    | HTTP/1.1 404 Not Found (text/html) |
| 398 | 115.202697908 | 127.0.0.1 | 127.0.0.1   | HTTP     | 490    | GET / HTTP/1.1                     |
| 399 | 115.202953094 | 127.0.0.1 | 127.0.0.1   | HTTP     | 254    | HTTP/1.1 304 Not Modified          |

Frame 399: 254 bytes on wire (2032 bits), 254 bytes captured (2032 bits) on interface lo, id 0

Ethernet II, Src: 00:00:00:00:00:00 (00:00:00:00:00:00), Dst: 00:00:00:00:00:00 (00:00:00:00:00:00)

Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1

Transmission Control Protocol, Src Port: 80, Dst Port: 56102, Seq: 637, Ack: 1126, Len: 188

Hypertext Transfer Protocol

HTTP/1.1 304 Not Modified

[Expert Info (Chat/Sequence): HTTP/1.1 304 Not Modified]

[HTTP/1.1 304 Not Modified]

[Severity level: Chat]

[Group: Sequence]

Response Version: HTTP/1.1

Status Code: 304

[Status Code Description: Not Modified]

Response Phrase: Not Modified

Server: nginx/1.18.0 (Ubuntu)

Date: Fri, 03 Oct 2025 15:09:24 GMT

Last-Modified: Fri, 03 Oct 2025 11:22:05 GMT

Connection: keep-alive

ETag: "680fb109-15"

[HTTP response 3/3]

[Time since request: 0.000255186 seconds]

[Prev request in frame: 395]

[Prev response in frame: 396]

[Request in frame: 398]

[Request URL: http://ifernersays.com/]

0000 00 00 00 00 00 00 00 00 00 00 00 00 08 00 45 00 .....E-

0010 00 f9 22 a3 48 00 40 00 19 63 7f 00 00 01 7f 00 ..." @ c-----

0020 00 01 00 50 0b 26 cc 42 7f c1 01 e0 94 7c 80 10 ...P & B .....|..

0030 02 00 fe e4 00 00 01 01 08 0a b1 22 64 2b b1 22 ..... "d+-"

0040 64 2b 48 54 54 50 2f 31 2e 31 20 33 30 34 20 4e d+HTTP/1 .1 304 N

0050 6f 74 20 40 6f 64 69 66 69 65 64 00 0a 53 65 72 ot Modif ied -Ser

0060 76 65 72 3a 20 6e 67 69 6e 78 2f 31 2e 31 38 2e ver: ngi nx/1.18.

Hypertext Transfer Protocol: Protocol

Packets: 530 · Displayed: 6 (1.1%)

Profile: Default

**4. Explique os pacotes HTTP que capturou, o seu significado e o seu conteúdo.**

1º pacote: Pedido HTTP GET da página iternerays.com ao servidor nginx.

2º pacote: Resposta do servidor com a página (o campo Line-based text data com o código do index.html da página), com código 200 e mensagem de Status OK.

3º pacote: pedido de um ícone/imagem ao servidor.

4º pacote: resposta com código 404 e status Not Found, a imagem não foi encontrada no servidor virtual.

5º pacote: Novo pedido HTTP GET da página ao servidor nginx, aquando do refresh da tab onde se acedeu primeiramente ao domínio iternerays.com. O campo If-Modified-Since do pacote indica a data do último pedido à página, e que se a página tiver sofrido alterações no servidor desde que o cliente fez o último pedido, para devolver a página atualizada.

6º pacote: resposta com código 304 e status Not Modified. Isto porque a página não sofreu alterações desde o último pedido da página ao servidor virtual nginx.